

HealthTech Knowledge Base AI Assistant

API Documentation (Week 1 Deliverable)

1. API Overview

System Name: HealthTech Knowledge Base AI Assistant

API Version: v1

Base Endpoint

Development:

<http://localhost:5000/api/v1>

Production:

<https://my-domain.com/api/v1>

Authentication

The API uses JWT-based authentication.

Authentication Header:

`Authorization: Bearer <JWT_TOKEN>`

2. Authentication APIs

POST /auth/login

Description

Authenticates users and generates a JWT access token.

Priority: P1

Access

Public

Request Parameters

- Email
- Password

Response

Returns:

- JWT token
- User information
- User role

Example response information:

- User ID
- Name
- Email
- Role

Roles Supported

- ADMIN
- EDITOR
- VIEWER

POST /auth/register

Description

Creates a new system user account.

Priority: P2

Access

Admin only

Request Data

Required:

- Name
- Email

- Password
- Role

Response

Returns:

- Created user details
- Account creation status

GET /auth/me

Description

Returns the currently authenticated user's profile.

Priority: P2

Access

Authenticated users

Response

Returns:

- User ID
- Name
- Email
- Role
- Account timestamps

POST /auth/logout

Description

Logs out the authenticated user and records the activity.

Priority: P2

Access

Authenticated users

3. Article Management APIs

GET /articles

Description

Retrieves all knowledge base articles.

Supports:

- Pagination
- Filtering
- Category filtering
- Article type filtering
- Status filtering

Priority: P1

Access

Authenticated users

Query Parameters

page

- Page number

limit

- Number of articles per page

status

- Published, Draft, Archived

type

- FAQ, Guide, Documentation

categoryId

- Filter by category

Response

Returns:

- Article list
- Article details
- Category
- Tags
- Author information
- Feedback count

GET /articles/:slug

Description

Retrieves a single article using its unique slug.

Priority: P1

Access

Public

Response

Returns:

- Article title
- Content
- Category
- Tags
- Author
- Views
- Related articles

POST /articles

Description

Creates a new knowledge base article.

Priority: P1

Access

EDITOR and ADMIN

Request Data

Includes:

- Title
- Content
- Article type
- Category
- Product
- Tags

Response

Returns:

- Creation status
- Article information

PUT /articles/:id

Description

Updates an existing article.

Priority: P2

Access

EDITOR and ADMIN

Request Data

Can update:

- Title
- Content
- Type
- Category
- Tags

Response

Returns updated article details.

PUT /articles/publish/:id

Description

Publishes an article and makes it available publicly.

Priority: P2

Access

ADMIN

Response

Returns:

- Published status
- Publication date

DELETE /articles/:id

Description

Deletes or archives an article.

Priority: P2

Access

ADMIN

Response

Returns deletion confirmation.

4. Category APIs

GET /categories

Description

Retrieves knowledge base categories.

Includes:

- Category name
- Category ID
- Category hierarchy

Priority: P1

Access

Public

Response

Returns available categories.

5. Search APIs

GET /search

Description

Searches the knowledge base using semantic and keyword search.

Supports:

- AI vector similarity search
- Full text search
- Filtering

Priority: P2

Access

Authenticated users

Query Parameters

q

- Search keyword

category

- Filter by category

type

- Filter by article type

product

- Filter by product

page

- Pagination

limit

- Results limit

Response

Returns:

- Matching articles
- Similarity score
- Article metadata

6. AI Chatbot APIs

POST /chat

Description

Main communication endpoint for the external HMIS chatbot widget.

Allows users to ask healthcare system questions.

Priority: P1

Access

Public / External Widget

Request Data

Includes:

- User question
- Chat session ID

Processing Flow

HMIS Chatbot Widget

↓

Chat API

↓

Express Backend

↓

Knowledge Base Database

↓

Vector Search (pgvector)

↓

AI Response Generation

Response

Returns:

- AI generated answer
- Session ID
- Source articles used

GET /chat/history/:sessionId

Description

Retrieves previous chat messages.

Priority: P2

Access

Authenticated users

Response

Returns:

- User questions
- AI answers
- Chat timestamps

POST /chat/feedback/:messageId

Description

Allows users to rate chatbot responses.

Priority: P2

Request Data

Includes:

- Helpful / Not helpful

Response

Returns feedback status.

7. Feedback API

POST /articles/:id/feedback

Description

Allows users to submit article ratings and comments.

Priority: P2

Access

Authenticated users

Request Data

Includes:

- Rating
- Comment

Response

Returns:

- Feedback status
- Updated rating statistics

8. Administration APIs

GET /admin/dashboard

Description

Provides administrative analytics.

Priority: P3

Access

ADMIN only

Returns

Dashboard information:

- Total article views
- Search activity
- Popular articles
- Usage statistics

9. Security Features

The API implements:

- JWT Authentication
- Role-Based Access Control
- Password encryption
- Request rate limiting
- CORS protection
- Secure HTTP headers
- Audit logging

10. API Priority Matrix

Endpoint	Priority
POST /auth/login	P1
GET /articles	P1
GET /articles/:slug	P1
POST /articles	P1
GET /categories	P1
POST /chat	P1
PUT /articles/:id	P2
DELETE /articles/:id	P2
PUT /articles/publish/:id	P2
GET /search	P2
POST /feedback	P2
GET /admin/dashboard	P3